Deep Learning - Week 12

1. What is the primary purpose of the attention mechanism in neural networks?

   (a) To reduce the size of the input data
   (b) To increase the complexity of the model
   (c) To eliminate the need for recurrent connections
   (d) To focus on specific parts of the input sequence

   Correct Answer: (d)
   Solution: The attention mechanism allows the model to weigh and prioritize different parts of the input sequence dynamically, giving more focus to the most relevant portions when making predictions or generating outputs. This is particularly useful in tasks like machine translation and image captioning, where different parts of the input may have varying levels of importance.

2. Which of the following are the benefits of using attention mechanisms in neural networks?

   (a) Improved handling of long-range dependencies
   (b) Enhanced interpretability of model predictions
   (c) Ability to handle variable-length input sequences
   (d) Reduction in model complexity

   Correct Answer: (a),(b),(c)
   Solution: Attention mechanisms help maintain long-range dependencies in input sequences, unlike traditional RNNs.
   Attention weights can sometimes be interpreted to understand which parts of the input influenced the model's decisions.
   Attention is inherently flexible to variable-length input sequences, especially in tasks like machine translation.

3. If we make the vocabulary for an encoder-decoder model using the given sentence. What will be the size of our vocabulary?
   Sentence: Attention mechanisms dynamically identify critical input components, enhancing contextual understanding and boosting performance

   (a) 13
   (b) 14
   (c) 15
   (d) 16

   Correct Answer: (c)
   Solution: There are 13 unique words in our vocabulary. We will add two additional words $<GO>$ and $<STOP>$ to the vocabulary. Hence the size of the vocabulary will be 15.

4. We are performing the task of *Machine Translation* using an encoder-decoder model. Choose the equation representing the *Encoder* model.

   (a) $s_0 = CNN(x_i)$
   (b) $s_0 = RNN(s_{t-1}, e(\hat{y}_{t-1}))$
   (c) $s_0 = RNN(x_{it})$
   (d) $s_0 = RNN(h_{t-1}, x_{it})$

   Correct Answer: (d)
   Solution: The encoder in an encoder-decoder model processes the input sequence by recursively updating its hidden state using the current input x_{it} and the previous hidden state h_{t-1}. This is expressed by the equation:

$$s_0 = RNN(h_{t-1}, x_{it})$$

5. Which of the following attention mechanisms is most commonly used in the Transformer model architecture?

   (a) Additive attention
   (b) Dot product attention
   (c) Multiplicative attention
   (d) None of the above

   Correct Answer: (b)
   Solution: In the Transformer model architecture, the most commonly used attention mechanism is **dot-product attention** (also referred to as scaled dot-product attention). This method calculates the attention scores by taking the dot product of the query and key vectors, which allows for efficient computation and parallelization.

   While additive attention and multiplicative attention are also valid attention mechanisms, they are not the primary focus in the Transformer architecture.

6. Which of the following is NOT a component of the attention mechanism?

   (a) Decoder
   (b) Key
   (c) Value
   (d) Query
   (e) Encoder

   Correct Answer: (a),(e)
   Solution: In the attention mechanism, the only components are the Query, Key, and Value. The encoder and the decoder are parts of the overall architecture that use attention but are not components of the attention mechanism itself.

7. In a hierarchical attention network, what are the two primary levels of attention?

(a) Character-level and word-level
(b) Word-level and sentence-level
(c) Sentence-level and document-level
(d) Paragraph-level and document-level

Correct Answer: (b)
Solution: Character-level and word-level: While character-level attention can be used in some NLP tasks, it's not typically one of the primary levels in a hierarchical attention network. Word-level and sentence-level: This is the correct answer. In a typical hierarchical attention network:

Word-level attention helps to identify important words within each sentence. Sentence-level attention then works on these sentence representations to identify important sentences within the document.

Sentence-level and document-level: While sentence-level attention is correct, document-level typically refers to the overall output rather than a separate level of attention. Paragraph-level and document-level: Paragraph-level attention is not commonly used as one of the primary levels in standard hierarchical attention networks.

Therefore, the correct answer is:

Word-level and sentence-level

This structure allows the network to build a hierarchical representation of the document, first by attending to important words to create sentence embeddings, and then by attending to important sentences to create the document embedding. This mimics the natural structure of many text documents and allows the model to capture both local (word-level) and global (sentence-level) contextual information.

8. Which of the following are the advantages of using attention mechanisms in encoder-decoder models?

(a) Reduced computational complexity
(b) Ability to handle variable-length input sequences
(c) Improved gradient flow during training
(d) Automatic feature selection
(e) Reduced memory requirements

Correct Answer: (b),(c),(d)
Solution: Explanation: B) Attention allows handling of variable-length inputs
C) Attention provides shorter paths for gradient flow
D) Attention automatically selects relevant features
A and E are incorrect as attention often increases computational and memory requirements.

9. In the encoder-decoder architecture with attention, where is the context vector typically computed?

(a) In the encoder

(b) In the decoder
(c) Between the encoder and decoder
(d) After the decoder

Correct Answer: (c)
Solution: In the encoder-decoder architecture with attention, the context vector is typically computed between the encoder and decoder. The attention mechanism uses the encoder's hidden states and calculates attention weights that highlight relevant parts of the input. The weighted sum of the encoder's hidden states is then used to form the context vector, which is passed to the decoder to assist in generating the output sequence.

10. Which of the following output functions is most commonly used in the decoder of an encoder-decoder model for translation tasks?

    (a) Softmax
    (b) Sigmoid
    (c) ReLU
    (d) Tanh

    Correct Answer: (a)
    Solution: In an encoder-decoder model for translation, the decoder's output layer typically produces a probability distribution over the target vocabulary, and Softmax is used to convert raw scores (logits) into probabilities that sum to 1.

    The Softmax function is applied after the decoder's output passes through a linear layer. It converts the raw scores (logits) into a probability distribution over the target vocabulary. Each value in the output represents the probability of a specific word being the next token in the sequence. This is essential in translation tasks, as it allows the model to select the most likely word at each step of the sequence generation.